

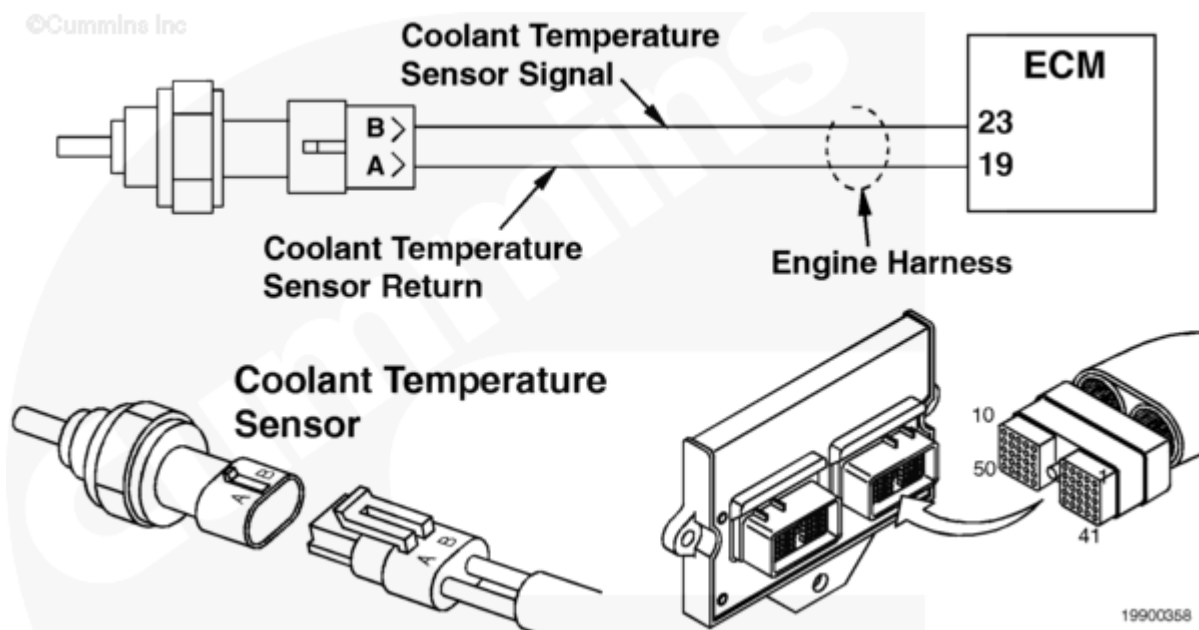
Fault Code: 144

Coolant Temperature Sensor Circuit

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 144 PID(P): P110 SPN: 110 FMI: 3 Lamp: Yellow SRT:	More than 4.95 VDC detected at the coolant temperature signal pin 23 of the engine harness.	Possible white smoke. Engine protection for coolant temperature is disabled.



Coolant Temperature Sensor Circuit

Circuit Description

The coolant temperature sensor is used by the electronic control module (ECM) to monitor the temperature of the engine coolant. The coolant temperature is used by the ECM for the engine protection system, timing, and fueling control.

The ECM monitors the voltage on pin 23. The ECM expects to see the voltage vary between 0.5 and 4.5 VDC. If the voltage is above 4.95 VDC, then the ECM will log Fault Code 144.

Voltage above 4.95 VDC on pin 23 can be caused by opens in the signal or return wires, or voltage short circuits to the signal or return wires, or a failed open sensor.

Component Location

The coolant temperature sensor is located on the left side of the engine in the thermostat housing.

Shoptalk

The resistance of the sensor varies with the temperature. The reading you observe will compare to the following table if the sensor is functioning properly.

Temperature (°C)	Temperature [°F]	Resistance (ohms)
0	32	30k to 36k
25	77	9k to 11k
50	122	3k to 4k
75	167	1350 to 1500
100	212	600 to 675

Refer to Troubleshooting Fault Code t05-144 (</qs3/pubsys2/xml/en/procedures/87/87-t05-144.html>)